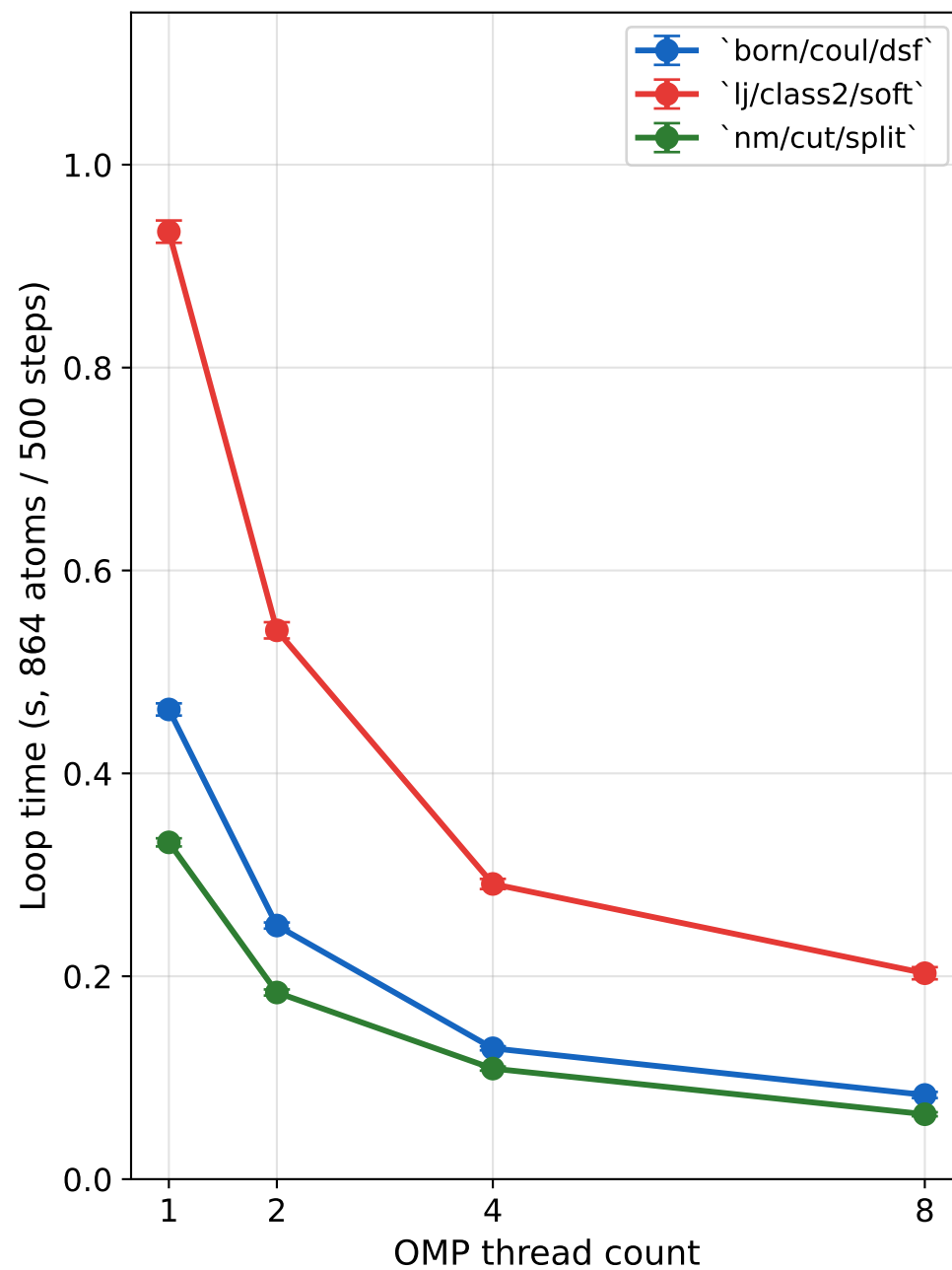
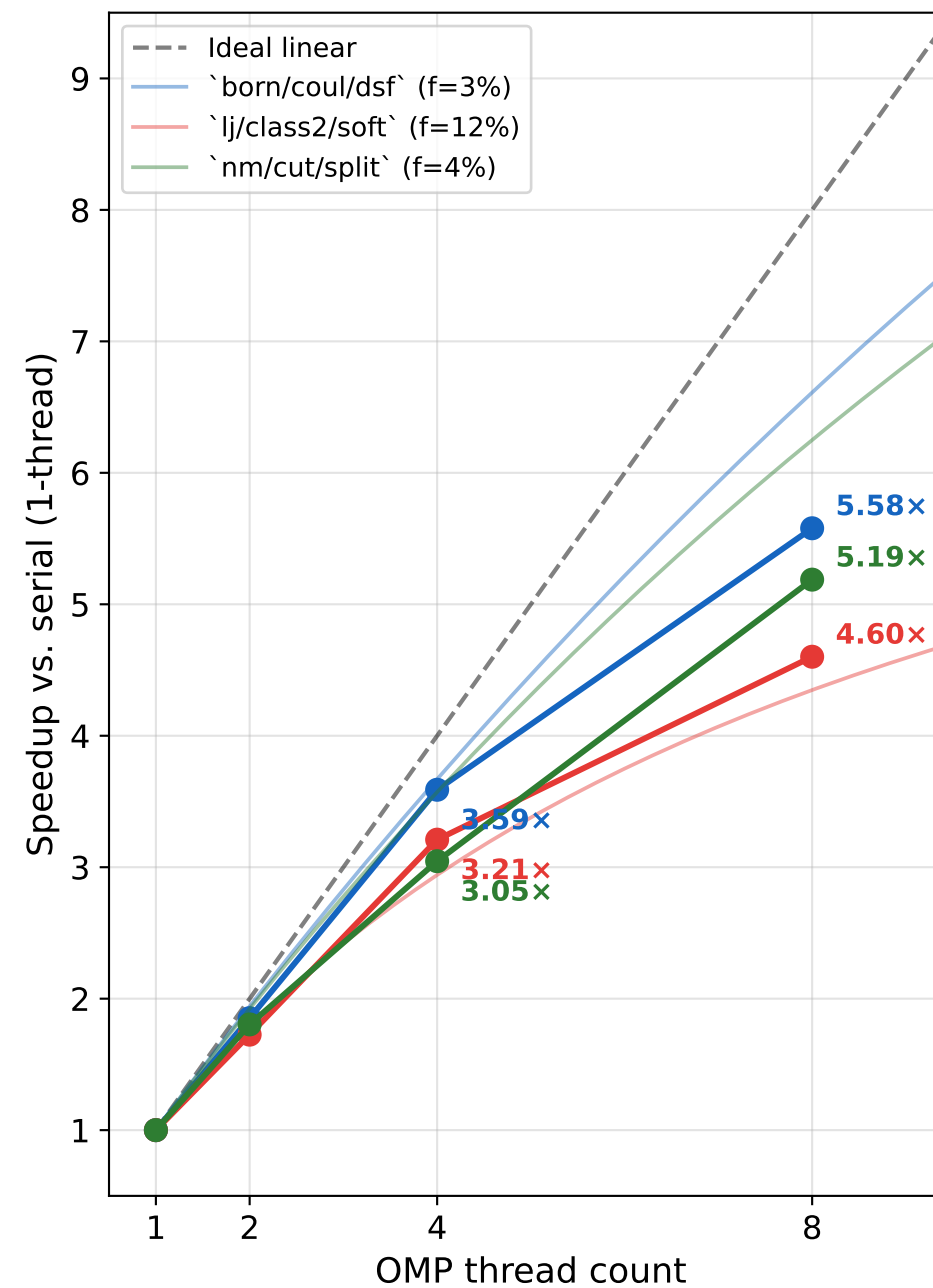


**Figure S1 · Full OMP scaling benchmarks for newly ported pair styles
(864-atom FCC lattice, 500 NVE steps, N=3 runs, $\pm 1\sigma$ error bars)**

(a) Loop time vs. thread count
(N=3 runs, error bars $\pm 1\sigma$)



(b) Speedup and Amdahl's law fit
(f = serial fraction)



(c) Pair-time fraction vs. thread count
(Amdahl limit from non-pair overhead)

